

App Development Project Report

App Dev Project Report

1. Student Details

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About Me: I am a standalone student at IIT Madras bs degree programme currently in the second term of diploma .

2. Project Details

Project Title: Placement Portal

Problem Statement:

Institutes need efficient systems to manage campus recruitment activities involving companies, students, and placement drives. Currently, many institutes rely on spreadsheets, emails, or manual processes, which makes it difficult to manage company approvals, track student applications, avoid duplicate registrations, and maintain placement records.

You are required to build a Placement Portal Application web application that allows Admin (Institute), Company, and students to interact with the system based on their roles.

Approach:

We will develop a role-based web portal with secure authentication, enabling companies to register and post jobs, students to apply, and admins to approve companies, manage drives, track applications. Replacing manual processes with a centralized, efficient digital system.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing SQLAlchemy model definitions, used for the css of the **home page**, using it to tackle and debug errors for higher proficiency.

The extent of AI/LLM usage is around **10–15%**.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
Flask-Login	User authentication and session management
Werkzeug	For hashing and password check
Flask-SQLAlchemy	Object Relational Mapper for SQLite database
HTML+Jinja2+Bootstrap 5	Frontend + Framework + templating
CSS	Minimally used(inline only)
SQLite	Lightweight local database for storing user data
JS	For managing the flash time of flask flashes (3 lines of code used)

5. Database Schema / ER Diagram

Tables:

1- users: id, email, password, role, is_active, is_approved, created_at

2-student_profiles: user_id, first_name, last_name, cgpa, department_name, graduation_year, resume_path

3-company_profiles: user_id, description, company_name, hr_number, website_url, approval_status

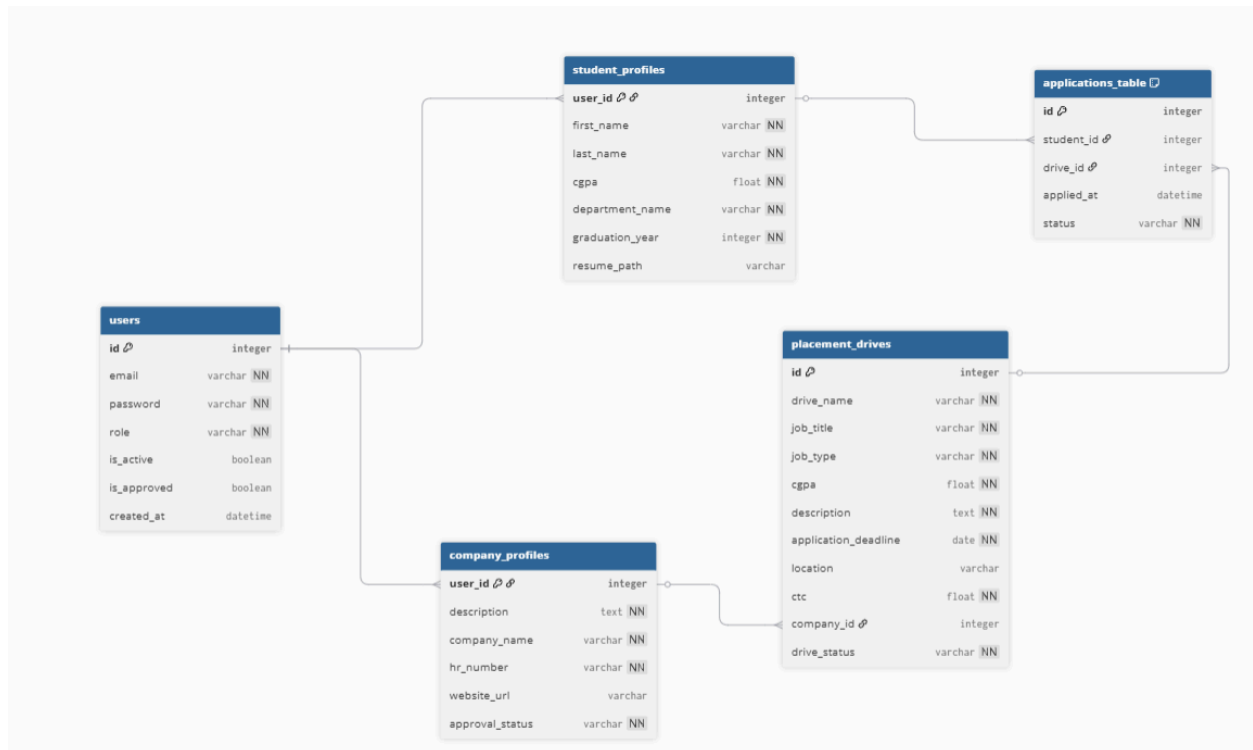
4-placement_drives: id, drive_name, job_title, job_type, cgpa, description, application_deadline, location, ctc, company_id, drive_status

5-applications_table:id, student_id, drive_id, applied_at, status

Relationships:

- One-to-One: users → student_profiles
- One-to-One: users company_profiles
- One-to-Many: company_profiles placement_drives
- One-to-Many: placement_drives → applications_table
- One-to-Many: student_profiles → applications_table

Diagram from db.io



Architecture Overview:

- **app.py** – main Flask application entry point and routing
- **models** – database models using SQLAlchemy
- **/templates** – Jinja2 HTML templates
- **/static** – CSS, and resume files

Implemented Features:

1. **Role-based authentication** system ensures Admin, Company, and Student access is strictly controlled, allowing secure login, approval workflows, and restricted dashboards based on user roles and permissions.
2. **Company approval mechanism** prevents unauthorized recruiters by requiring Admin verification before accessing dashboards or creating placement drives, ensuring only legitimate organizations participate in recruitment processes.

3. **Placement drive management** allows companies to create, update, and control job postings with eligibility, deadlines, and status, while Admin approval ensures quality and consistency of opportunities.
 4. **The student application system** enables applying to drives, tracking application status, and maintaining history, while preventing duplicate applications to ensure fair participation and accurate recruitment records.
 5. **The centralized admin dashboard** provides complete control over students, companies, drives, and applications, including search, approval, rejection, and blacklisting for efficient placement process management.
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8. Video Presentation

Drive Link:

 Placement_Portal.mp4
